HEAPLINE · FEASIBILITY ASSESSMENT

HeapLine feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First assessment, written for the kit 1.9.1 upgrade

HeapLine feasibility assessment

On paper HeapLine is feasible as a kit a local workshop can build from stock steel, plywood and rope for an estimated USD 827 a site plus USD 547 for each shared capstan set; the open questions are field facts that only trials can settle: how hard slide debris resists a probe, and how far a shared capstan can safely reach. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	A pumped friction capstan, sectional probes, linked crawl boards and a sheet stretcher, all sized on paper; every concept option modelled to D3 (A 347 parts, B 236, C 217) with no interference	High	Probe reach in stiff debris is within 3 % of the push two people can give (HPL-STM-001 S12); the shear pin spread leaves 2 % margin to the 5 kN pull (S6)	HPL-STM-001, HPL-CAL-001
Materials	Steel tube and bar, exterior plywood, treated timber, UV-stabilised HDPE, acetal, polyester rope and webbing; no exotic material	High	Coating life in a hot, damp shed to assess; rope bend ratio on the roller is 4.3 (S9)	BOM, HPL-STM-001 S9
Manufacturing and assembly	Cut, drill and weld from stock; the making sketches and 15 steps are in the build plan; a village fabricator with a small welder can make the capstan	Medium	Weld quality and fit-up to prove in the first build; profile-cut ratchets and pawls need a laser or plasma cutter, or hand cutting	HPL-BLD-001, `cad/fidelity.json`
Power and controls	None: people supply the power; no electrics beyond AA lamps	High	Lamps use alkaline cells stored out of the lamps	BOM line 25
Supply of parts	Common stock sizes; bought parts are rope, slings, shackles, shovels, padlocks, hinges and lamps	Medium	Rated slings and shackles with threaded pins to source (HPL-REG-001 G3, G4); a reused crate must be found at each site	BOM, HPL-REG-001
Cost	USD 827 a site against USD 850 (PR-009); USD 1,374 for the host site against the USD 1,800 value-engineering target	High	All prices indicative, not quotes; a bought steel job box would take a site over USD 850	HPL-CAL-001 G1 to G3
Safety and regulatory	No whole-product regulation found; a safety case rests on the overload pin, holding pawls, sound anchors, stop rule and lookout	Medium	Whether a made stretcher or a shared capstan may be used in law is to assess (HPL-REG-001 G9, G10); never certified rescue equipment	HPL-RSK-001, HPL-REG-001
Field performance	Unknown until trials: debris resistance, drill times, how long a shared capstan takes to arrive from 500 m	Low	No published data on probe resistance in dumpsite slide debris; paper estimates bracket it with a loosened and a stiffer case	HPL-CAL-001 B1, F4

Ratings: High, Medium, Low.

Overall: feasible on paper at TRL 3. Trials with a co-design partner (TRL 4, not started) would settle the Low-rated field aspect.